

WASP-12b

R-0269

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_241025 · created 2026-07-17T12:48:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460608.9205 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.011
Transit depth (Rp/Rs)^2	1.14 +/- 0.23 [%]
Orbital Inclination (inc)	80.51 +/- 1.97
Airmass coefficient 1 (a1)	1.151 +/- 0.011
Airmass coefficient 2 (a2)	-0.1307 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	0.96 %
Best Comparison Star	None
Optimal Aperture	4.65
Optimal Annulus	13.19
Transit Duration (day)	0.1157 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.011 vs 0.1178 (deviation 0.95σ)
Duration fit vs published	0.1157 d vs 0.125 d (deviation 1.41σ)
Mid-time O–C	0.9 min
Depth SNR	9.4
Residual scatter	0.96 %

Light curve

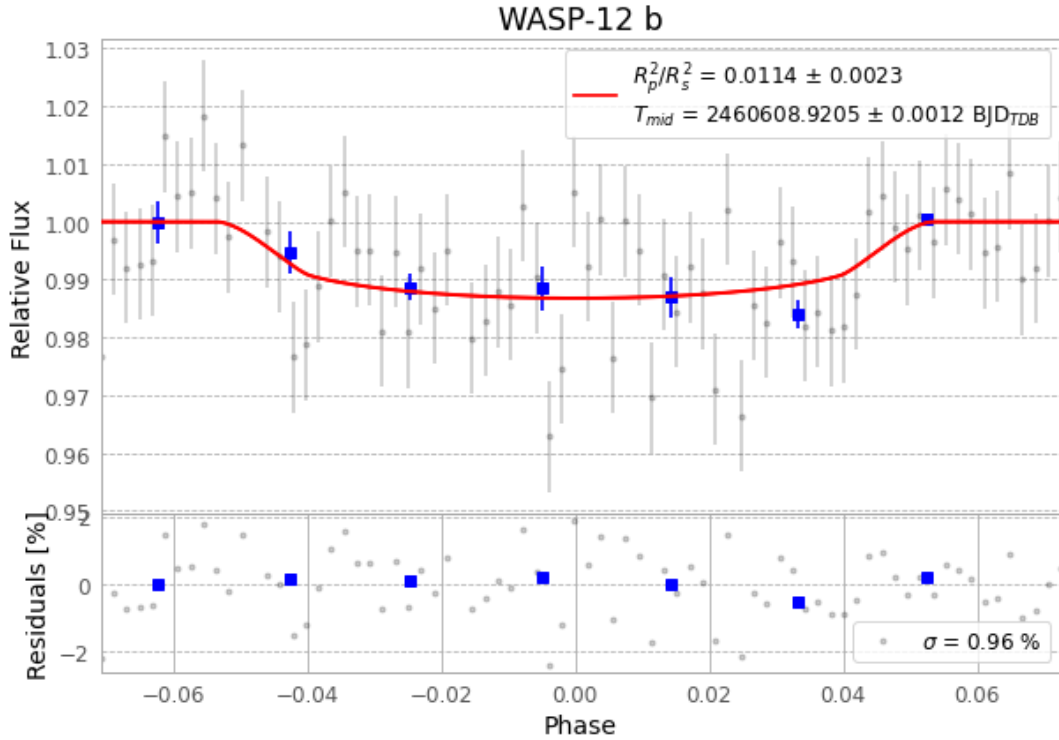


Figure 1 — FinalLightCurve_WASP-12 b_2024-10-25.png (SHA-256 0f631dbc133c4b4a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [377, 179], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 98c3b5c4032d0196...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1cb2214

Data provenance

76 FITS frames (50 MB), WASP-12241025080915.FITS → WASP-12241025115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241025.log (028d5a3611aa...), FinalLightCurve_WASP-12 b_2024-10-25.png (0f631dbc133c...), FinalLightCurve_WASP-12 b_2024-10-25.csv (1eb14ea8c297...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 12:48Z.